ELTPH401 –PNEUMATICS AND HYDRAULICS SYSTEM INSTALLATION

DATE: .../.../2025
PERIOD: .....

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM I EXAMINATIONS: 2025-2026

QUESTION PAPER

SECTOR: Energy

TRADE NAME: Electrical Technology

RQF LEVEL: 4

MODULE TITLE: PNEUMATICS AND HYDRAULICS SYSTEM INSTALLATION

MARKS: **...../100**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: 26 questions, all are **Compulsory**. **55 marks**
Section **B**: Five 5 questions, choose and answer any three (3). **30 marks**
Section **C**: Two 2 questions, choose and answer any one (1) **15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

SECTION A

MULTIPLE CHOICE*/20MARKS*

1. Pneumatics primarily uses which medium for power transmission?
A. Water
B. Oil
C. Compressed air
D. Steam

2. Which of the following is *NOT* a primary function of a hydraulic reservoir?
A. Dissipating heat
B. Air separation
C. Fluid compression
D. Contamination settling

3. A **4/3 valve** has:
A. 4 ports, 3 positions
B. 3 ports, 4 positions
C. 4 paths, 3 actuators
D. 3 ports, 4 switching signals

4. A hydraulic hose must be selected based on:
A. Colour and appearance
B. Pressure rating, temperature, fluid compatibility
C. The brand name
D. Weight only

5. Which pump type delivers **constant flow regardless of pressure**?
A. Centrifugal pump
B. Gear pump
C. Rotodynamic pump
D. Dynamic pump

6. A **flow control valve** is used to:
A. Change fluid type
B. Control actuator speed
C. Store hydraulic fluid
D. Increase compressor horsepower

7. Which fitting is commonly used for **quick assembly without tools**?
A. Threaded fitting
B. Barbed fitting
C. Push-to-connect fitting
D. Compression fitting

8. In indirect control, the actuator movement is triggered by:
A. Direct push button
B. Manual valve
C. Relay or solenoid control
D. Mechanical steering

9. A flow sensor that uses **heat transfer** to measure flow is known as:
A. Electromagnetic
B. Thermal mass
C. Turbine
D. Ultrasonic

10. Which device stores pressurized fluid energy in a hydraulic system?
A. Pump
B. Accumulator
C. Cylinder
D. Motor

TRUE OR FALSE QUESTIONS*/20MARKS*

1. Hydraulics uses a compressible fluid to transmit power.

2. Pneumatic systems usually operate at lower pressures than hydraulic systems.

3. Filters in hydraulic systems remove contaminants to protect components.

4. A double-acting cylinder has only one port.

5. Cavitation is a common cause of pump failure.

6. All hose fittings in pneumatic systems require crimping.

7. Flow control valves are used to regulate actuator speed.

8. Hydraulic flanges are not suitable for high-pressure applications.

9. Workplace ergonomics includes the correct use of furniture and layout design.

10. Indirect control uses electrical or logic elements to control pneumatic actuators.

✓ 5 FILL IN THE BLANK QUESTIONS*/10MARKS*

1. The device used to measure fluid pressure in hydraulic and pneumatic systems is called a _____.

2. A _____ is used to reduce and maintain a set output pressure in a circuit.

3. Pneumatic actuators convert compressed air energy into _____ or _____ motion.

4. A _____ pump traps a fixed volume of fluid and forces it through the discharge line.

5. Threaded connections require _____ tape or sealant to prevent leaks.

MATCHING QUESTIONS*/5MARKS*

Match COLUMN A with COLUMN B

COLUMN A	COLUMN B
1. Accumulator	A. Stores pressurized fluid energy
2. Compressor	B. Converts mechanical power into air pressure
3. Flow sensor	C. Measures rate of fluid movement
4. Reservoir	D. Stores and conditions hydraulic fluid
5. Coupling	E. Connects two shafts for power transmission

SECTION B. Short Answer (5 Questions) */30MARKS*

1. Describe what the arrows inside a DCV symbol indicate.
2. What does a pilot line in pneumatic symbols represent?
3. Explain the graphical difference between a fixed and adjustable flow control valve.
4. Draw or describe the symbol for a pressure relief valve.
5. Why do hydraulic symbols not show internal construction details?

SECTION C. Open-Ended (2 Questions) */15MARKS*

1. Explain how to interpret a 4/3 hydraulic directional control valve symbol, including ports, positions, and internal arrows.
2. Discuss the importance of standardization of hydraulic and pneumatic symbols in industry.